Project code

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

// Pins
#define VOLTAGE_PIN A0
#define CURRENT_PIN A1
#define TEMP_PIN A2

#define MOSFET_PIN 6
#define BUZZER_PIN 7

#define GREEN_LED 8
#define YELLOW_LED 9
#define RED_LED 10

// Variables
float voltage = 0;
float current = 0;
float temperature = 0;

// Calibration
float voltageFactor = 5.0;
float currentOffset = 2.5;
float sensitivity = 0.185;
```

```
String statusMsg = "";

void setup() {
  Serial.begin(9600);

  lcd.init();
  lcd.backlight();

  pinMode(MOSFET_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);

  pinMode(GREEN_LED, OUTPUT);
  pinMode(YELLOW_LED, OUTPUT);
  pinMode(RED_LED, OUTPUT);

  digitalWrite(MOSFET_PIN, HIGH);

  lcd.setCursor(0,0);
  lcd.print("EV BMS System");
  delay(2000);
  lcd.clear();
}

void loop() {

  // 🔋 Voltage
  int vRaw = analogRead(VOLTAGE_PIN);
```

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voltage = vRaw * (5.0 / 1023.0) * voltageFactor;

// ⚡ Current
int cRaw = analogRead(CURRENT_PIN);
float sensorVoltage = (cRaw * 5.0) / 1023.0;
current = (sensorVoltage - currentOffset) / sensitivity;
if (abs(current) < 0.05) current = 0;

// 🌡️ Temperature
int tRaw = analogRead(TEMP_PIN);
temperature = tRaw * (5.0 / 1023.0) * 100;

// 🚨 FAULT
if (voltage > 12.6 || voltage < 9.0 || temperature > 50 || current > 2.0) {

  digitalWrite(MOSFET_PIN, LOW);
  digitalWrite(RED_LED, HIGH);
  digitalWrite(YELLOW_LED, LOW);
  digitalWrite(GREEN_LED, LOW);
  digitalWrite(BUZZER_PIN, HIGH);

  statusMsg = "FAULT!";
}

// ⚠️ WARNING
else if (voltage > 11.0) {

  digitalWrite(MOSFET_PIN, HIGH);
```

```
  digitalWrite(YELLOW_LED, HIGH);
  digitalWrite(RED_LED, LOW);
  digitalWrite(GREEN_LED, LOW);
  digitalWrite(BUZZER_PIN, LOW);

  statusMsg = "WARNING";
}

// ✅ NORMAL
else {

  digitalWrite(MOSFET_PIN, HIGH);
  digitalWrite(GREEN_LED, HIGH);
  digitalWrite(YELLOW_LED, LOW);
  digitalWrite(RED_LED, LOW);
  digitalWrite(BUZZER_PIN, LOW);

  statusMsg = "NORMAL";
}

// 📟 LCD DISPLAY (NO FLICKER)
lcd.setCursor(0,0);
lcd.print("V:");
lcd.print(voltage,1);
lcd.print(" T:");
lcd.print(temperature,1);

lcd.setCursor(0,1);
```

```
  lcd.print("I:");
  lcd.print(current,1);
  lcd.print(" ");
  lcd.print(statusMsg);

  // Clear extra characters (important)
  lcd.print("   ");

  // Serial monitor
  Serial.print("V:");
  Serial.print(voltage);
  Serial.print(" I:");
  Serial.print(current);
  Serial.print(" T:");
  Serial.println(temperature);

  delay(1000);
}
```